

Lab 2 - Configuring Basic Single-Area OSPFv2

Answer Sheet

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Part 2: Configure and Verify OSPF Routing

Step 1: Verify OSPF neighbors and routing information.

b.

What command would you use to only see the OSPF routes in the routing table?

____ show ip route ospf _____

Part 4: Configure OSPF Passive Interfaces

- g. Re-issue the **show ip route** and **show ip ospf neighbor** commands on R1 and R3, and look for a route to the 192.168.2.0/24 network.

What interface is R3 using to route to the 192.168.2.0/24 network? ____ s0/0/0 ____

What is the accumulated cost metric for the 192.168.2.0/24 network on R3? ____ 129 ____

Does R2 show up as an OSPF neighbor on R1? __ Yes __

Does R2 show up as an OSPF neighbor on R3? __ No __

What does this information tell you?

____ R3 is currently linking to R2 via R1 instead of directly connect to R2, the traffic is using a longer path to link between R3 and R2 _____

- h. Change interface S0/0/1 on R2 to allow it to advertise OSPF routes. Record the commands used below.

____ router ospf 1 _____

____ network 192.168.23.0 0.0.0.3 area 0 _____

- i. Re-issue the **show ip route** command on R3.

What interface is R3 using to route to the 192.168.2.0/24 network? ____ s0/0/1 ____

What is the accumulated cost metric for the 192.168.2.0/24 network on R3 now and how is this calculated?

____ 64 (R3 s0/0/1 cost)+1 (R2 G0/0 cost) _____

Is R2 listed as an OSPF neighbor to R3? ____ Yes ____

Part 5: Change OSPF Metrics

Step 1: Change the reference bandwidth on the routers.

- a. To reset the reference-bandwidth back to its default value, issue the **auto-cost reference-bandwidth 100** command on all three routers.

```
R1(config)# router ospf 1
```

```
R1(config-router)# auto-cost reference-bandwidth 100
```

```
% OSPF: Reference bandwidth is changed.
```

```
Please ensure reference bandwidth is consistent across all routers.
```

Why would you want to change the OSPF default reference-bandwidth?

To ensure the calculation for the cost remains accurate. It can prevent OSPF from automatically choosing a faster path or Ethernet.

Step 2: Change the bandwidth for an interface.

g.

Explain how the costs to the 192.168.3.0/24 and 192.168.23.0/30 networks from R1 were calculated.

_For 192.168.3.0, cost = 781 + 1 = 782 (R1 s0/0/1 cost + R3 G0/0 cost) _

_For 192.168.23.0, cost = 781 + 64 = 845 (R1 cost + R3 s0/0/1 cost) _

- i. Issue the **bandwidth 128** command on all remaining serial interfaces in the topology.

What is the new accumulated cost to the 192.168.23.0/24 network on R1? Why?

_1562, because cost = 781 + 781 (R1+ R3) _

Step 3: Change the route cost.

c.

Explain why the route to the 192.168.3.0/24 network on R1 is now going through R2?

Since the cost of the direct connection to R3 is higher than going through R2, the path is changed.

Reflection

1. Why is it important to control the router ID assignment when using the OSPF protocol?
___ Because it will be easier to troubleshoot and the network will be more stable. ___
___ If no ID is assigned, OSPF will choose the highest IP on the loopback or physical interface. If the interface is down, the RID will change, and OSPF has to restart and recalculate. Using an ID can prevent it and make the routers more specific. _
2. Why is the DR/BDR election process not a concern in this lab?
___ Because it only occurs on multi-access networks, e.g., Ethernet. In this lab, routers are connected via Point-to-Point Serial links. No election is needed, since every link has only two devices.
3. Why would you want to set an OSPF interface to passive?
___ It can increase security and reduce unnecessary traffic. It will prevent the routers from sending or receiving OSPF "Hello" packets on the port. ___
4. What aspect of OSPF makes the protocol hierarchical and permits the creation of very scalable networks?
___ The use of Areas makes OSPF hierarchical. OSPF will divide a large network into multiple areas, limiting the propagation of Link-State Advertisements (LSAs), and keep routing table calculations local to each area. It can reduce the CPU and memory load on each router, allowing the network to scale. ___
5. What single OSPF router configuration command allows the assignment of OSPF area 0 to all interfaces in the range 10.0.0.0 to 10.255.255.255?
___ network 10.0.0.0 0.255.255.255 area 0 ___
6. A router has three routes to the same network: one route from a RIP with a metric of 4, another from OSPF with a metric of 3053092, and another from EIGRP with a metric of 4039043. Which of the three routes will be used for the routing decision?
___ EIGRP will be used, since its administrative distance (AD) is the smallest (90). The metric is no longer important if the ADs are not the same among those three routes ___